

P.V.S.S. D.A.V. PUBLIC SCHOOL

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STUDENT'S PROFILE:-

Participants:-

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TITLE: - 'Mann Ki Awaaz' - The Virtual Guide Within You

Primary Theme: - AATMANIRBHAR BHARAT

Idea Category: - Future-Ready Models

1. NATURE OF WORK

This project is an interdisciplinary initiative combining Artificial Intelligence, Internet of Things, and Behavioral Psychology. The work involves:

- ✚ Hardware engineering to create a wearable prototype using Raspberry Pi 5, camera, and audio components
- ✚ Software development integrating computer vision, speech processing, and advanced AI models
- ✚ AI prompt engineering to ensure contextually appropriate ethical guidance and mental health support
- ✚ User experience design for seamless, non-intrusive daily integration

2. RATIONALE

"Mann Ki Awaaz" addresses two critical modern challenges:

- ✚ Erosion of Civic Sense: Increasing public negligence, littering, and traffic violations
- ✚ Mental Health Epidemic: Rising loneliness, stress, and anxiety with limited support systems

The project leverages technology not for automation but for humanization, providing immediate, contextual feedback for both external behavior and internal well-being.

3. SCIENTIFIC PRINCIPLE

The system operates on core scientific principles:

- ✚ Computer Vision: OpenCV and machine learning for object recognition and action detection
- ✚ Natural Language Processing: Advanced LLMs (GPT-4) for contextual understanding and response generation
- ✚ Cognitive Behavioral Theory: Immediate feedback mechanism to reshape behavior and thought patterns
- ✚ Edge Computing: Local processing on Raspberry Pi for reduced latency and bandwidth usage

4. MATERIAL REQUIRED

HARDWARE:

Raspberry Pi 5, Pi Camera Module

Bone Conduction Headphones/Speaker

Power Bank, MicroSD Card

3D-Printed Wearable Case

SOFTWARE:

Python 3.9+, OpenCV Library

SpeechRecognition, Python TTS

OpenAI GPT-4 API/Open-Source LLM

Raspberry Pi OS

5. PROCEDURE

 Hardware Assembly: Connect PiCam, audio components to Raspberry Pi

 Software Setup: Install OS and necessary Python libraries

 Module Development:

 Vision Module: Object and action detection scripts

 Audio Module: Speech-to-text and text-to-speech functions

 AI Integration: API bridge for contextual processing

 AI Training: Develop dual-mode prompts for civic guidance and mental health support

 Testing & Refinement: Integrate modules and test in controlled environments

- ✚ Prototype Finalization: House components in wearable enclosure (e.g., a small box that clips to a shirt or integrated into glasses frames)

6. ADVANTAGES

- ✚ Dual Benefit: Addresses both social ethics and mental wellness
- ✚ Real-Time Intervention: Provides immediate reinforcement for habit formation
- ✚ Accessibility: 24/7 support at low cost compared to human therapy
- ✚ Personalization: AI adapts to individual user patterns over time
- ✚ Proactive Prevention: Helps de-escalate stressful situations before they worsen
- ✚ Stigma Reduction: Private, judgment-free channel for self-improvement

7. SCOPE OF LARGE SCALE IMPLEMENTATION

- ✚ Educational Sector: Digital life skills coach for students
- ✚ Corporate Wellness: Employee mental health and collaboration support
- ✚ Public Health: Municipal programs for civic sense and mental health
- ✚ Healthcare: Supplementary tool for therapists and patient monitoring
- ✚ Tourism: Cultural etiquette guide and travel companion
- ✚ Future Tech: Integration into AR glasses, smart helmets, and vehicle interfaces